

Multiple Choice (10 marks)

1. For what value of b is the line $y = 10x$ tangent to the curve $y = e^{(bx)}$ at some point in the xy -plane?
 - (a) $10/e$
 - (b) 10
 - (c) $10e$
 - (d) e^{10}
2. How many structurally distinct Abelian groups have order 72?
 - (a) 4
 - (b) 6
 - (c) 8
 - (d) 9
3. Which of the following is an equation of the line tangent to the graph of $y = x + e^x$ at $x = 0$?
 - (a) $y = 2x + 1$
 - (b) $y = 2x$
 - (c) $y = x + 2$
 - (d) $y = x + 1$
4. The element $(4, 2)$ of $\mathbb{Z}_{12} \times \mathbb{Z}_8$ has order
 - (a) 4
 - (b) 8
 - (c) 12
 - (d) 6
5. Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^3 + cx^2 + 1)$ is a field.
 - (a) 0
 - (b) 2
 - (c) 1
 - (d) 3

True / False (10 marks)

Answer True or False

6. True or False: There exists a group of order 102 that does not have a nontrivial proper normal subgroup.
7. True or False: The order of the factor group $\mathbb{Z}_6/\langle 3 \rangle$ is 3.
8. True or False: Every compact space in a metric space is complete.
9. True or False: The set of all n th roots of unity under multiplication of complex numbers forms an abelian group.
10. True or False: The degree of the field extension $\mathbb{Q}(\sqrt{2} + \sqrt{3})$ over $\mathbb{Q}$ is 4.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the minimum value of

$$f(x) = \sqrt{5x^2 + 2x\sqrt{5} + 1} + x\sqrt{5},$$

over all x for which $f(x)$ is defined.

12. A polynomial with integer coefficients is of the form

$$3x^3 + a_2x^2 + a_1x - 6 = 0.$$

Enter all the possible integer roots of this polynomial, separated by commas.

End of Paper